

CHECKLY ATTENDANCE SYSTEM

Technical Documentation

May 2026

1 INTRODUCTION

1.1 Purpose

Checkly Attendance System is a mobile and web-based application that is built using Microsoft Power Apps, Microsoft SharePoint and Power Automate. This system was designed to improve the employee attendance management process. Aside from that, this system was built to replace the old attendance method that required manual tracking such as electronic punch cards that are currently used in many organizations. Therefore, with the help of this new system, it can help to improve the attendance tracking processes by having features that can record check ins and out, overview insights, generate reports and so on.

1.2 Scope

This system serves three types of user roles which are:

- i. **Staff** – able to record their daily attendance and view their personal dashboard as well as their reports.
- ii. **Head of Department (HOD)** – able to record their daily attendance, view their personal dashboard and reports as well as view department dashboard and reports.
- iii. **Admin** – able to record their daily attendance, view their personal dashboard and reports as well as view admin full dashboard and admin full reports including user management where user are able to view, create, update or delete the data.

This system also includes key functions such as:

- i. Daily check in and check out with work mode selection (On-Site or Work From Home)
- ii. Location-based validation for Work Mode (On-Site)
- iii. Weekly quota enforcement in Log Attendance (3 days On-Site, 2 days WFH)
- iv. Public holiday management
- v. Automated absent marking at the end of each working date if no data has been taken
- vi. CSV export functions in report page

2 System Overview

2.1 High level Architecture

The system is built on three Microsoft platforms working together:

Component	Platform	Role
Frontend App	Microsoft Power Apps	Used to design user interface for all roles
Data Storage	Microsoft SharePoint	Used to stores attendance, user data, and public holidays
Automation	Microsoft Power Automate	Used to schedule flows, trigger email and import data

2.2 Data Flow

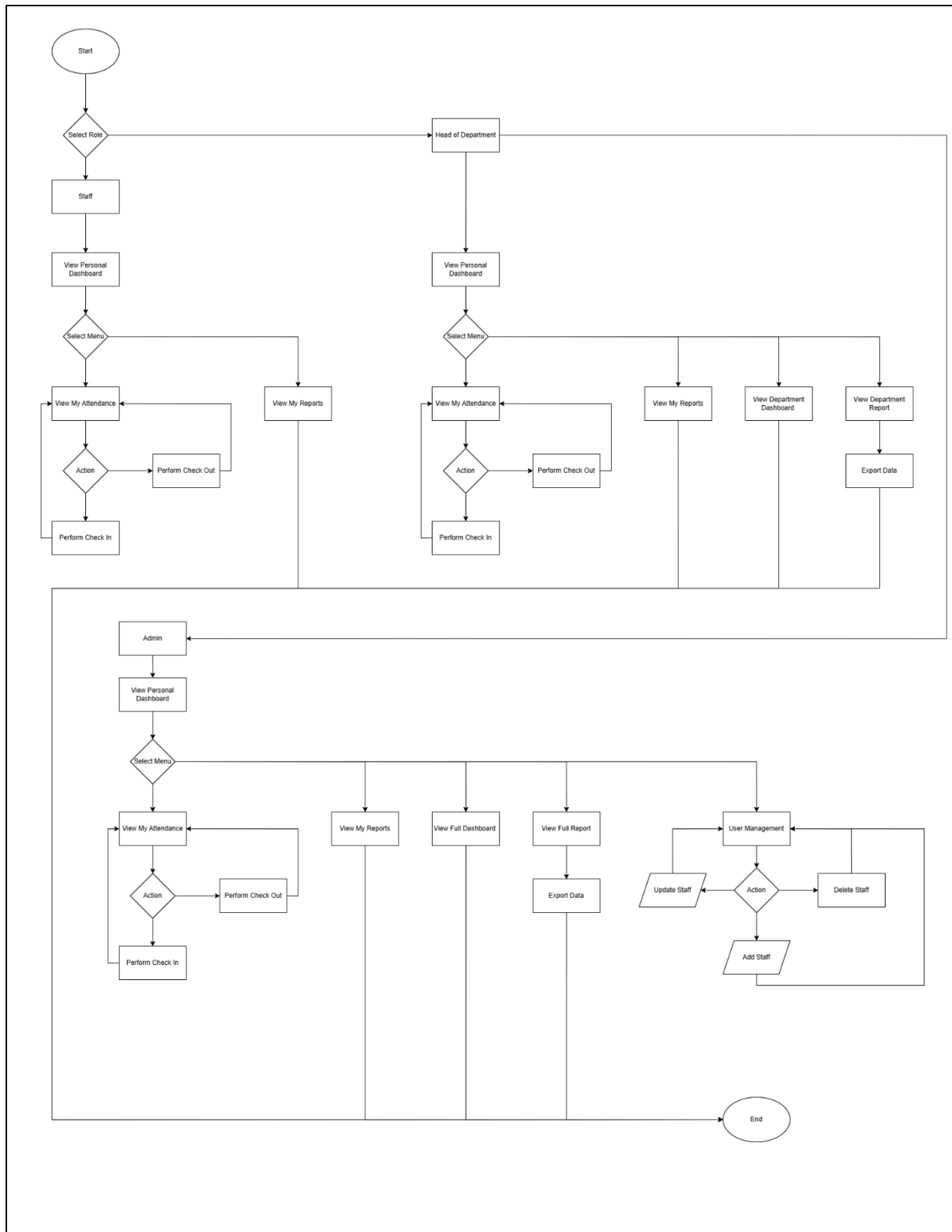


Figure 2.1 Flowchart of Checkly System

2.3 Office Coordinates

Setting	Value
Office Longitude	101.4128674
Office Latitude	2.8680942
Check in Distance Limit	10KM
Work Day Start	8:40 AM

3 Features and Functions

3.1 Attendance Check In (LogAttendanceScreen)

Roles that have access: **Staff/HOD/Admin**

- i. Work mode must be selected: On-Site or Work From Home
- ii. On-Site check in requires user to be within 10KM of the office
- iii. Check in after 8:40 AM is automatically marked as Late Arrival
- iv. Weekly quota: 3 days On-Site and 2 days WFH per week
- v. Weekly On-Site and WFH remaining days
- vi. Average Working Hours calculation for the day
- vii. Duplication for check in on the same day is blocked
- viii. GPS coordinations (Latitude, Longitude) are recorded for On-Site check ins

3.2 Personal Dashboard (MyDashboardScreen)

Roles that have access: **Staff/HOD/Admin**

- i. Working Mode Status
- ii. Status filtering for Weekly Trend chart
- iii. Total days present (current month)
- iv. Late arrival count (current month)
- v. Average working hours (current month)
- vi. Attendance Rate Pie Chart (current month)
- vii. Weekly Trends Bar Chart (current week)

3.3 Personal Report (MyReportScreen)

Roles that have access: **Staff/HOD/Admin**

- i. Status and Month filtering for Data table
- ii. Data table showing personal attendance records

3.4 HOD Department Dashboard (DeptDashboardScreen)

Roles that have access: **HOD**

- i. Total staff in department
- ii. Average Attendance Rate (% On Time)
- iii. Total Late Arrival
- iv. Weekly Late Arrival Bar Chart (current month)
- v. Weekly On Time Arrival Bar Chart (current month)

3.5 HOD Department Report (DeptRecordScreen)

Roles that have access: **HOD**

- i. Staff name, Status, Date filtering for Data table
- ii. Data table showing department attendance records
- iii. Export data as CSV file to HOD logged in email

3.6 Admin Full Dashboard (FullDashboardScreen)

Roles that have access: **Admin**

- i. Department filtering for all KPI cards, charts and table
- ii. Total staff registered in an organization
- iii. Average Attendance Rate (% On Time) (current month)
- iv. Total Staff Late (current month)
- v. Monthly Attendance Breakdown Bar Chart (current year)
- vi. Weekly Late Arrival Bar Chart (current month)
- vii. Weekly On Time Arrival Bar Chart (current month)
- viii. Data table for Staff Summary details

3.7 Admin Full Report (FullReportScreen)

Roles that have access: ***Admin***

- i. Department, Staff name, Status, Work mode, From and To filtering for Data table
- ii. Data table showing full organization attendance records
- iii. Export data as CSV file to Admin logged in email

3.8 User Management (UserMngScreen)

Roles that have access: ***Admin***

- i. Tab: Staff List
 - a) Staff name filtering for Data table
 - b) Add, Edit, Delete staff records
- ii. Tab: Public Holiday
 - a) Month filtering for Data table
 - b) Add, Edit, Delete public holidays records

3.9 Automation Features

- i. Auto Mark Absent
 - a) Runs daily at 5:30 PM, skip weekends and public holidays and Mark absent if no attendance was recorded for the day after 5:30 PM
- ii. Check out Reminder
 - a) Runs daily at 5:15 PM, sends email reminder to logged in users who have checked in for the day

4 Data Source

4.1 SharePoint Lists

i. UsersList

Column	Type	Description
FullName	Single line of text	Employee full name
Email	Single line of text	Employee email address
Role	Choice	Admin/HOD/Staff
Department	Single line of text	Employee department

ii. AttendanceList

Column	Type	Description
UserEmail	Single line of text	Employee email address
Department	Single line of text	Employee department
Date	Date only	Attendance date
CheckInTime	Date and Time	Time of check in
CheckOutTime	Date and Time	Time of check out
Status	Choice	In Time / Late Arrival / Absent
workHours	Number	Total working hours for the day
WorkMode	Choice	On-Site / Work From Home
Latitude	Number	GPS latitude at check in
Longitude	Number	GPS longitude at check out

iii. PublicHolidayList

Column	Type	Description
Title	Single line of text	Name of the public holiday
HolidayDate	Date only	Date of the public holiday

5 User Guide

5.1 Accessing the App

- i. Open Microsoft Power Apps on browser or mobile
- ii. Sign in with your Microsoft account
- iii. Select Checkly Attendance System from your apps list

5.2 Staff – How to Check In

Step 1:

- i. Open the app and navigate directly to Personal Dashboard screen
- ii. View KPI cards and charts of personal attendance records
- iii. Click filter button to filter the chart by Status (All, On Time, Late Arrival, Absent)
- iv. Click on Reset button to reset the filtering

Step 2:

- i. Navigate to Log Attendance Screen
- ii. Select your Work Mode (On-Site or Work From Home).
- iii. Click the Check In button.
- iv. If On-Site mode, ensure you are within 20KM from the office. The app will verify your location automatically.
- v. A success notification will appear with your check-in time.

Important rules:

- i. You can only check in once per day
- ii. Check-in after 8:40 AM is recorded as Late Arrival
- iii. Maximum 3 On-Site days and 2 WFH days per week
- iv. You will receive a reminder email at 5:15 PM for check out reminder
- v. If you do not check in at all, you will be marked Absent automatically at 5:30 PM

Step 3

- i. Navigate to Personal Report screen
- ii. Click filter button to filter the data table using Status and month filtering
- iii. Click on Reset button to reset the filtering
- iv. View the data table of personal attendance records

5.3 HOD – Viewing Department Dashboard

Step 1

- i. Navigate to Department Dashboard screen
- ii. View all the KPI cards and charts based on current month

Step 2

- i. Navigate to Department Report screen
- ii. Click filter button to filter data table based on Staff name, Status and date filtering
- iii. Click on Reset button to reset the filtering
- iv. View the table of department attendance records
- v. Click the Export Data to export full department attendance report

5.4 Admin – Full Dashboard

Step 1

- i. Navigate to Full Dashboard screen
- ii. Click filter to filter KPI cards, charts and table based on Department name filtering
- iii. View KPI cards, charts and data table
- iv. Click on Reset button to reset the filtering

Step 2

- i. Navigate to Full Report screen
- ii. Click filter to filter data table based on Department name, Staff name, Status, Work Mode, Date From and To filtering
- iii. Click on Reset button to reset the filtering

- iv. Click the Export Data to export full organization employee's attendance report

Step 3

- i. Navigate to User Management screen
- ii. View Staff List tab
- iii. Search Staff name using the filtering section
- iv. Click on Add User to add new staff details
- v. View the Staff records in Data table view
- vi. Edit or delete the staff details

Step 4

- i. Select Public Holiday tab
- ii. View Public Holiday tab
- iii. Search Public Holiday using month filtering
- iv. Click on Add Holiday to add new public holiday details
- v. View the Public holiday in Data table view
- vi. Edit or delete the public holiday

6 Power Automate Flows

Flow Name	Trigger	Purpose
Export Attendance Report	PowerApps (manual)	Exports HOD department report as CSV email
ExportFullAttendanceReport	PowerApps (manual)	Exports full attendance data as CSV for Admin
AutoMarkAbsent	Scheduled — 5:30 PM daily	Marks absent for users with no attendance record for the day
CheckOutReminder	Scheduled — 5:15 PM daily	Send reminder emails to users who have not checked out
Import Users to UsersList	Power Automate (manual)	Import data in excel file into the SharePoint List (UsersList)
Import Attendance to AttendanceList	Power Automate (manual)	Import data in excel file into the SharePoint (AttendanceList)

** All scheduled flows will automatically skip weekends (Saturday and Sunday) and public holidays registered in the PublicHolidayList.

7 Maintenance & Troubleshooting

7.1 Common Issues

Issue	Possible Cause	Solution
Check-in button not working	Formula error and invalid arguments	Check OnSelect formula in Power Apps
Distance check always failing	Browser GPS inaccuracy	Set larger radius distance
Absent not being marked	Flow not running or holiday conflict	Check AutoMarkAbsent run history in Power Automate
CSV export is empty	Filter query returning no data	Check Filter Array step in flow run history
Dashboard showing old month data	Month filter missing in formula	Ensure Month(Date) = Month(Today()) filter is applied
Reminder email not received	User already checked out or flow failed	Check CheckOutReminder run history
Filtering errors	Formula errors	Check OnSelect formula in Power Apps
Log attendance quota	Does not apply formula for limit quota	Set rules and validation for the Check In button
Absent marked on public holiday and weekend	Flow condition not working	Check Condition step in AutoMarkAbsent
Unable to import data into SharePoint List	There are two tables inside the excel file	Check the table name and call the right name inside the Power Automate
Time imported into SharePoint is different with inside the excel file	Due to the UTC timestamps	Using formula to convert to Local Time Zone
Missing data in data table report in Power Apps	Due to Pagination limit	Update the pagination limit to 1000
Missing imported data in SharePoint List	Due to pagination limit	Update the pagination to 1000

8 Appendices

Checkly project

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Intern

Project start: **Sun, 4/19/2026**

Display week: **1**

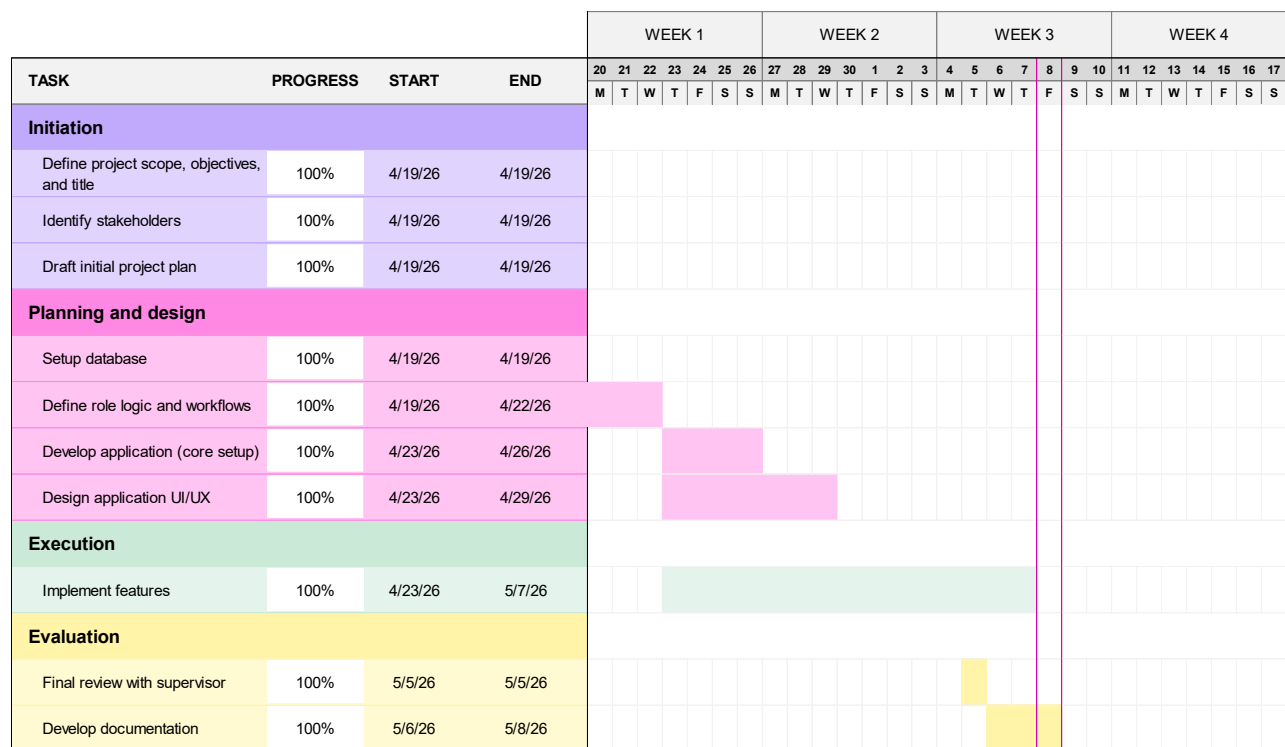


Figure 8.1 Gantt Chart for Checkly System